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Structure-Guided Design of a Broadly Cross-Reactive Multivalent Group A Streptococcal Vaccine

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Abstract

The M protein of group A streptococci (Strep A) is a major virulence determinant and protective antigen. The N-terminal region of the M protein is variable in sequence, defines the M/emm type, and contains epitopes that elicit opsonic antibodies that protect animals from challenge infections. Although there are >200 M types of Strep A, there is now evidence that structurally related M proteins can be grouped into clusters and that immunity may be cluster-specific in addition to M type-specific. This observation has led to recent studies of structure-based design of multivalent M peptide vaccines to select peptides predicted to cross-react with heterologous M types to improve vaccine coverage. In the current study, we have applied a refined series of peptide structural algorithms to predict immunological cross-reactivity among 117 N-terminal M peptides representing the most prevalent M types of Strep A. Based on the results of the structural analyses, in combination with global M type prevalence data, we constructed a 32-valent vaccine containing 19 cross-reactive vaccine candidates predicted to cross-react with 37 heterologous M peptides to which were added 13 type-specific M peptides. The 4-protein recombinant vaccine was immunogenic in rabbits and elicited significant levels of antibodies against 31/32 (97%) vaccine

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Declaration of interests

Dr. Dale is the inventor on certain patents related to the design and development of vaccines to prevent group A streptococcal infections. The intellectual property is owned by the University of Tennessee Research Foundation.

All other authors declare no competing interests.

peptides and 28/37 (76%) peptides predicted to cross-react. The vaccine antisera also promoted opsonophagocytic killing of vaccine and cross-reactive M types of Strep A. Based on a recent analysis of M type prevalence of Strep A, the potential global coverage of the 32-valent vaccine is ~90%, ranging from 68% in Africa to 95% in North America. Our results indicate the utility of structure-based design that may be applied to future studies of broadly protective M peptide vaccines.

Keywords

Streptococcus pyogenes (*S. pyogenes*); M protein; peptide structures; vaccine development

1. Introduction

Group A *streptococcus* (Strep A) is a human-specific pathogen that is responsible for a significant disease burden worldwide [1]. Efforts to develop vaccines to prevent Strep A infections and their complications have been ongoing for many decades [2]. Many have focused on the M protein as a vaccine antigen, which is a major virulence determinant and protective antigen [3]. The variable sequence N-terminal region of M protein defines the M type of Strep A and contains epitopes that elicit antibodies with the greatest opsonic activity [4] that protect animals from challenge infections [5]. We have previously constructed recombinant multivalent M protein-based vaccines containing limited N-terminal M peptides linked in tandem representing as many as 30 different M types of Strep A [6, 7]. Clinical studies in adult volunteers have indicated that the vaccines are safe and immunogenic and elicited functional opsonic antibodies against the M types represented in the vaccines [8, 9].

One challenge in developing multivalent M peptide vaccines is the number of M/emm types (>200), each of which by definition expresses a surface M protein with a different N-terminal peptide sequence. Given the diversity of the epidemiology of Strep A infections, especially in populations at high risk for rheumatic heart disease (RHD) [10, 11], even complex multivalent vaccines may not achieve satisfactory levels of M type coverage. It has been widely held for many years that immunity to Strep A is type-specific and infection with one M type confers protection only against the infecting type. However, with the advent of *emm* sequencing, it has become evident that the majority of M types are structurally related and can be grouped into sequence-based clusters that share host protein binding characteristics [12]. This observation and our previous results showing that a 30-valent vaccine elicited antibodies that cross-opsonized multiple “non-vaccine” M types [6] prompted us to take a structure-based approach to the design of multivalent M peptide vaccines [13–15].

Our approach is based on the concept that Immunity to GAS infections is a combination of “cluster-specific” and “type-specific” immune responses that depend on the sequence-related conformational properties of the M epitopes concerned. Our hypothesis is that broadly protective vaccines can be developed using structure-guided design that takes full advantage of the structural relatedness of M peptides. The overall goal of the current study was to develop and validate a set of structural algorithms that would predict which M peptides are

most likely to elicit cross-opsonic antibodies against non-vaccine M types, thus increasing the potential emm type coverage of the vaccine.

2. Methods

2.1 Peptide structure-based algorithms used for vaccine design

Antibodies bind to epitopes in three dimensions. Our hypothesis is that 3D structure is more predictive of antibody binding than sequence alone. Therefore, similarity in epitope 3D structure between two sequence-related M peptides is hypothesized to predict antibody cross-reactivity. Here, we used a set of structural criteria shown to predict with high specificity the likelihood of eliciting significant levels of cross-reactive antibodies between pairs of sequence-related heterologous M peptides [13–15]. Previous studies have identified 117 sequence-related N-terminal M peptides from M types that are responsible for >90% of Strep A infections globally [14]. After assessing pairwise sequence identity of all 117 peptides, additional criteria considered the empirical heptad identity scores of the peptide pairs, the length of the predicted coiled-coil region and the degree of coiled-coil propensity. The empirical heptad score acts as a descriptor that captures the degree of sequence and structural similarity between a pair of coiled-coil peptides. The score makes use of the fact that coiled-coil structures are adopted by the majority of M-protein sequences which have structural repeats every seven residues. To determine the heptad score between peptide pairs, we established the heptad register with MARCOIL [16] and then calculated the pairwise heptad score as the sum of the weights associated with the sequence identity at each of the polar heptad positions as calculated by the Needleman Wunsch algorithm [17]. The hydrophobic core residues were not included as they were assumed to be less accessible for antibody binding. Each possible cross-reactive peptide-pair was further classified into four classes based on the length of the shortest predicted coiled-coil region of the peptide in the pair and the lowest coiled-coil propensity of the peptide in the pair. Peptides in Class 1 pairs had long coiled-coil regions (>35aa) and the lowest coiled-coil propensity of either peptide in the pair was 20%–50%, Class 2 pairs had peptides that each had long coiled-coil regions (>35aa) and high coiled-coil propensity (>50%). Peptide pairs in Class 3 had a minimum length of coiled-coil regions of 25–34 aa and moderate to high coiled-coil propensity (>25%). Class 4 pairs contained at least one peptide with a short coiled-coil region (<25 aa) and/or low coiled-coil probability (<20%). Class 4 pairs were not predicted to cross-react. Class 1 and Class 2 pairs were predicted to cross-react if the coiled-coil region sequence identity was >50% and the empirical heptad score was >10. Class 3 pairs were predicted to cross-react if the coiled-coil region sequence identity was >55% and the empirical heptad score was >13.5.

2.2 Construction, expression, and purification of vaccine proteins

Using the algorithms outlined above, in combination with global estimates of Strep A M type prevalence [17], 19 M peptides that were predicted to cross-react with one or more heterologous M types were selected to include in the multivalent vaccine. An additional 13 M peptides with type-specific aa sequences were included to maximize the potential Strep A coverage using global prevalence data. Four recombinant hybrid vaccine proteins were designed containing the 32 N-terminal M peptides (Fig. 1). Genes were synthesized

(GenScript, Piscataway, NJ) to encode the four vaccine proteins. In each gene, the coding sequence for the N-terminal peptide was reiterated on the 5' and 3' regions based on previous studies indicating that a duplicated peptide could function as a sacrificial peptide in the event that proteases preferentially degraded the N- or C-terminal regions [18]. The synthetic gene, which also contained a T7 promoter and a 3' His-tag, was ligated into pUC57, which was used to transform *E. coli* carrying the T7 RNA pol. The recombinant proteins were expressed and then purified by metal chelate chromatography, as previously reported in [18].

2.3 Immunization of rabbits

Three female New Zealand white rabbits (Charles River, Wilmington, MA, USA) were immunized intramuscularly with 400 µg of the vaccine (100 µg of each protein) mixed with squalene-in-water (Addavax, InvivoGen, San Diego, CA, USA) containing 25 µg of monophosphorylated lipid A (MPL, InvivoGen) at time 0, 4, and 8 weeks. A booster injection was given at 18 weeks and serum was obtained 2, 3, and 4 weeks after the final injection and pooled. The pooled immune serum from individual rabbits was used in all of the following antibody assays.

2.4 Peptide antibody assays

Pre-immune and immune rabbit antisera were assayed for antibody levels against synthetic M peptides by ELISA, as previously described [19]. ELISA titers were expressed as the inverse of the last dilution of antiserum that resulted in an O.D. of ≥ 0.15 . The 50 amino acid peptide antigens were custom synthesized by Peptide2Go (Manassas, VA, USA) using solid-phase peptide synthesis, desalted, and supplied at 70% purity.

2.5 HL-60 opsonophagocytic killing (OPK) assays

OPK assays were performed in quadruplicate using HL-60 cells as effector phagocytes and baby rabbit serum as a source of complement, as previously described [20]. Percent killing was calculated using the formula: $[(\text{total CFU with normal rabbit serum (NRS)}) - (\text{CFU with immune serum})] / (\text{total CFU with NRS}) \times 100$. Data were reported as the average percent killing in quadruplicate assays $\pm$ the standard deviation calculated after averaging the CFU in the four control samples.

2.6 Calculating potential vaccine coverage using epidemiologic data

The potential emm type coverage of the 32-valent vaccine was estimated using predicted and observed vaccine peptide specific and cross-reactive peptide antibody responses to the vaccine in immunized rabbits in combination with recently curated global Strep A surveillance data [17].

2.7 Institutional approvals

All animal procedures performed in this study were approved by the University of Tennessee Animal Care and Use Committee.

3. Results

3.1 Vaccine design

Applying the sequence and structure-based criteria for predicting peptide cross-reactivity to the hypervariable N-terminal 1–50 residues of 117 M proteins, we identified 69 cross-reactive M peptide pairs and 72 emm types that were predicted to cross-react with one or more emm types (Supplemental Data, Table 1). We initially selected the minimum number of M peptides that based on cross-reactivity would theoretically cover all 72 emm types which resulted in a 25-valent hypothetical vaccine that would cover the remaining 48 cross-reactive non-vaccine emm types. However, the predicted global coverage for these 73 emm types as calculated using the regional epidemiological data [17] was low and uneven across the different geographic regions. Therefore, we retained the top 19 cross-reactive vaccine candidates from the 72 cross-reactive emm types and added 13 type-specific M peptides based on their prevalence (Fig. 1 and Supplemental Data, Table 2). The 19 selected emm types were predicted to cross react with 37 heterologous emm types and, in combination with the 13 type-specific M peptides, provide theoretical M-type coverage of 90% globally.

3.2 Immunogenicity of the 32-valent vaccine

Three rabbits were immunized with the 32-valent vaccine formulated in squalene-in-water plus MPL. The vaccine was highly immunogenic and elicited significant levels of antibodies against the N-terminal synthetic peptides contained in the vaccine (Fig. 2). Maximum serum antibody titers in any of the three immunized rabbits ranged from 400 to 102,400. Antibody levels in at least one of three antisera increased by 8-fold or greater over pre-immune levels against 31 of 32 (97%) vaccine peptides. Of the 96 titers determined (3 rabbit antisera $\times$ 32 peptides), 80 (83%) demonstrated at least an 8-fold increase in antibody levels.

3.3 Cross-reactive M peptide antibodies elicited by the vaccine

The M peptides included in the 32-valent vaccine were predicted to cross-react with N-terminal M peptides of 37 non-vaccine emm types of Strep A (Fig. 1). At least 1 of the 3 immune rabbit sera contained significant levels of antibodies (≥ 8 -fold) against 28/37 (76%) non-vaccine M peptides (Fig. 3). Antibody levels in one or more of the three antisera increased by 4-fold over pre-immune levels against 34/37 (92%) non-vaccine peptides. To assess the accuracy of the structural algorithms used to select the cross-reactive peptide pairs, the immune sera were also assayed for cross-reactivity against the remaining 48/117 M peptides not predicted to share epitopes with the vaccine peptides. Antibody titers of ≥ 400 were observed with 3/48 peptides (M13, M15, and M96) (data not shown).

3.4 Predicted versus observed peptide cross-reactive antibodies and theoretical Strep A emm type vaccine coverage

To determine the potential coverage of the 32-valent vaccine, we combined the predicted or observed M peptide cross-reactive immune responses to the 32-valent vaccine together with the recent compilation of Strep A emm type prevalence in regions of the world [19]. The potential global coverage based on the M types included in the vaccine and predicted cross-reactive antibody responses was estimated to be 90% (Table 1). However, regional

coverage ranged from 68% in Africa and Oceania to 95% in North America. Applying observed antibody responses (≥ 8 -fold) resulted in 86% predicted emm type coverage in the world, with a range of 60% in Africa to 91% in Asia and the Pacific.

3.5 Vaccine-specific and cross-reactive OPK activity

Opsonophagocytic killing assays were performed to assess the functional activity of the 32-valent vaccine immune responses (Fig. 4). Twenty-nine of the 32 vaccine emm types and 29/37 emm types predicted to cross-react with the 32-valent vaccine antisera were assessed in HL-60 killing assays. Only emm types that met the pre-established criteria for growth in the control mixtures (≥ 4 generations) were included in the analysis. Assays with each M type were performed in quadruplicate with each of the three rabbit antisera and only the antiserum resulting in the highest mean killing value is reported. Killing of $\geq 30\%$ was observed for 21/29 vaccine emm types tested (Fig. 4). Some level of killing ($\geq 10\%$) was achieved with 28/29 vaccine emm types. The mean level of killing for all vaccine emm types tested was 46% (range 0–98%). Of the 29 cross-reactive emm types tested, 12 resulted in killing $\geq 30\%$ and the mean killing was 29% (range 0–81%).

4. Discussion

The search for safe, effective and affordable vaccines to prevent Strep A infections and their complications has been ongoing for decades. Obstacles to vaccine development include the complex biology of Strep A, the variability of the host response to infection, the possibility of autoimmunity triggered by vaccines, and the lack of a defined human immune correlate of protection. Currently there are two general approaches to vaccine formulation. One involves the identification of protective antigens that are shared by most or all M types of Strep A and the other is formulating multiple N-terminal M peptides in multivalent vaccines targeting the most prevalent emm types [2]. While M antibodies protect against infection in several animal models [5], the challenge has been to develop multivalent vaccines that would provide optimal coverage against the >200 emm types of Strep A.

In recent studies, we have taken a structure-based approach to capitalize on the immunological relatedness of M proteins [13–15]. The observation that the majority of M proteins are structurally related to others has led to the concept of M clusters [12]. Within a cluster, the sequence-related M proteins also share host protein binding characteristics. It is likely that the requirement to bind specific host proteins, which is associated with the virulence of different M types within a cluster, serves to constrain the structures which contributes to immunological cross-reactivity. We have previously tested this hypothesis by applying structure-based analyses to three different clusters of M proteins in an effort to minimize the total number of vaccine M peptides necessary to elicit functional immune responses (opsonophagocytic killing) against the non-vaccine emm types of Strep A in the cluster. The process is iterative and requires postulating structural determinants likely to predict cross-reactive antibody binding between pairs of M peptides, formulating vaccines containing the minimum number of peptides predicted to cross-react with all emm types in the cluster, testing the vaccines to empirically determine the cross-reactivity among the M

types of Strep A in the cluster, refining the structural algorithms based on the results, and redesigning new vaccines using refined structural algorithms (Fig. 5).

In the current study, we have further refined the suite of analytical tools used to predict which M peptides are most likely to elicit antibodies that cross-react with heterologous non-vaccine emm types of Strep A. Rather than starting with pre-defined clusters of M peptides, we applied the new set of structural analyses to 117 N-terminal M peptides from emm types of Strep A that are most prevalent in the world [14]. We used the criteria of sequence identity, coiled-coil length and propensity, plus heptad identity scores to predict the likelihood that two peptides would display immunological cross reactivity. For practical reasons related to the size of recombinant vaccine proteins, the number of potential cross-reactive peptides in the vaccine was down-selected based on the global and geographic prevalence of M types [17]. The addition of type-specific peptides to increase the potential coverage of the vaccine was likewise based on M type prevalence.

The 32-valent vaccine elicited antibodies that cross-reacted with 28/37 peptides (76%) predicted by the structural algorithms. When the antisera were assayed against the 48 M peptides that were not predicted to share epitopes, cross-reactions were observed against 3/48 (6%) of the remaining peptides. These results indicate favorable positive and negative predictive values of the structural criteria used to select the vaccine peptides. Improving the predictive power of the analyses will require additional iterations of the process using even more precise structural criteria. Co-crystallization studies with cross-reactive antibodies and peptides may shed light on new structural features involved in cross-reactive epitope conformations.

In this study, we used an HL-60 cell OPK assay to determine the functional activity of antibodies elicited by the 32-valent vaccine. We have previously shown that results of the HL-60 assay are generally comparable to those achieved with the classical Lancefield whole human blood assay [20]. However, the HL-60 assay also appears to be a less sensitive, yet more specific indicator of functional antibody activity compared to the whole blood assay because of the presence of naturally acquired Strep A antibodies in human blood. There is not an established correlation between the absolute level of bacterial killing promoted by immune sera and protection against infection. The establishment of an immune correlate will require advanced clinical trials designed to assess the efficacy of vaccines in preventing Strep A infections and simultaneous determinations of OPK activity, and potentially additional immune signatures such as T-cell responses and cytokine profiles. Nonetheless, in the current study we observed functional antibody activity against the majority of vaccine and non-vaccine M types tested.

The predicted global M type coverage of the 32-valent vaccine was estimated using a data set of M type prevalence of Strep A recently curated from the literature [17]. The results for some geographic regions and those of global coverage are likely skewed by the relative paucity of molecular epidemiologic studies in low- and middle-income regions of the world. Of the 77,068 emm types of Strep A included in the analysis, the vast majority were reported from North America (47%), Europe (29%), Asia-Pacific (12%), and Oceania (9%). The lower levels of coverage predicted in Africa, Arab countries, South

America, and Oceania are well below the 90% efficacy target stated in the Strep A Preferred Product Characteristics recently outlined by the WHO [21]. Nonetheless, coverage of even 60% of infections may eventually translate into a significant impact on the morbidity and mortality associated with Strep A infections and their complications. Continued efforts to identify structural algorithms that predict which M peptides are likely to cross-react with heterologous M types of Strep A should further enhance the potential coverage and efficacy of M protein-based vaccines, especially in regions of the world with populations at high risk for ARF and RHD.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- There is a global need for broadly protective group A streptococcal vaccines
- M protein-based subunit vaccines elicit M-type cross-reactive antibodies
- Structure-based design of multivalent vaccines improves potential coverage
- A 32-valent vaccine could prevent the majority of group A streptococcal infections

Protein 1	M1	M89	M18	M28	M74	M183	M187	M58	M1
Predicted cross-reactive M peptides	emm227 emm239	emm84 emm232				emm9			
Protein 2	M12	M78	M108	M66	M83	M87	M102	M82	M12
Predicted cross-reactive M peptides	emm229		emm70	emm90 emm104		emm79 emm209 emm103	emm114 emm124 emm175 emm77		
Protein 3	M3	M22	M11	M8	M5x2	M2x2	M178	M67	M3
Predicted cross-reactive M peptides		emm60 emm109	emm44 emm65 emm85	emm88 emm169		emm73	emm80 emm186 emm223		
Protein 4	M4	M49	M6x2	M43	M59	M76	M81	M75	M4
Predicted cross-reactive M peptides		emm118		emm53 emm123	emm63 emm94 emm177	emm92 emm106 emm27		emm48	

Fig. 1.

32-valent vaccine designed to elicit broadly cross-reactive antibodies against heterologous M peptides. The vaccine contained 19 M peptides predicted to elicit cross-reactive immunity to an additional 37 M types (listed under the respective vaccine peptide), plus 13 M peptides from prevalent M types that were not predicted to cross-react with others but were included to increase potential vaccine coverage. M67 and M11 were each predicted to elicit cross-reactive antibodies against M44, M65, and M85 peptides. M2, M5, and M6 peptides were 25 aa in length and repeated in tandem (x2) for a total length of 50 aa.

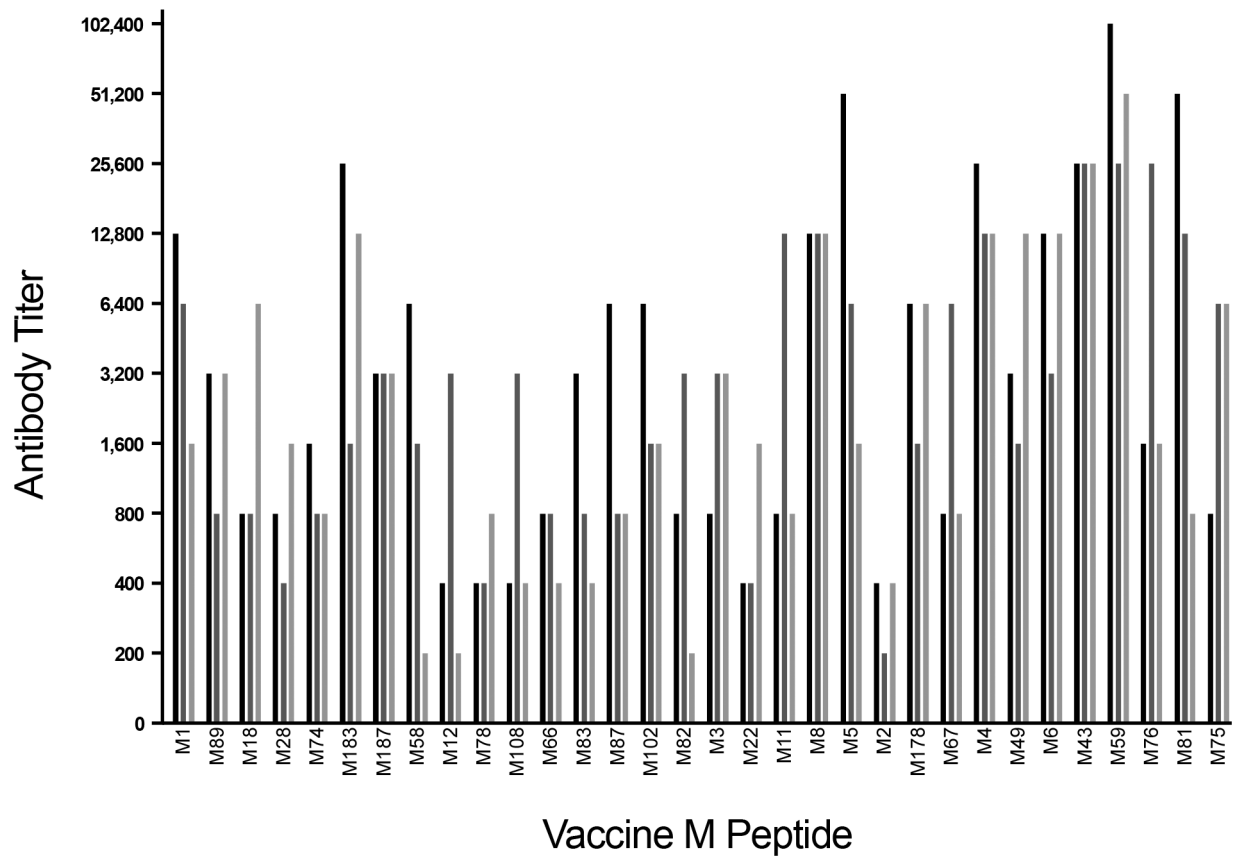
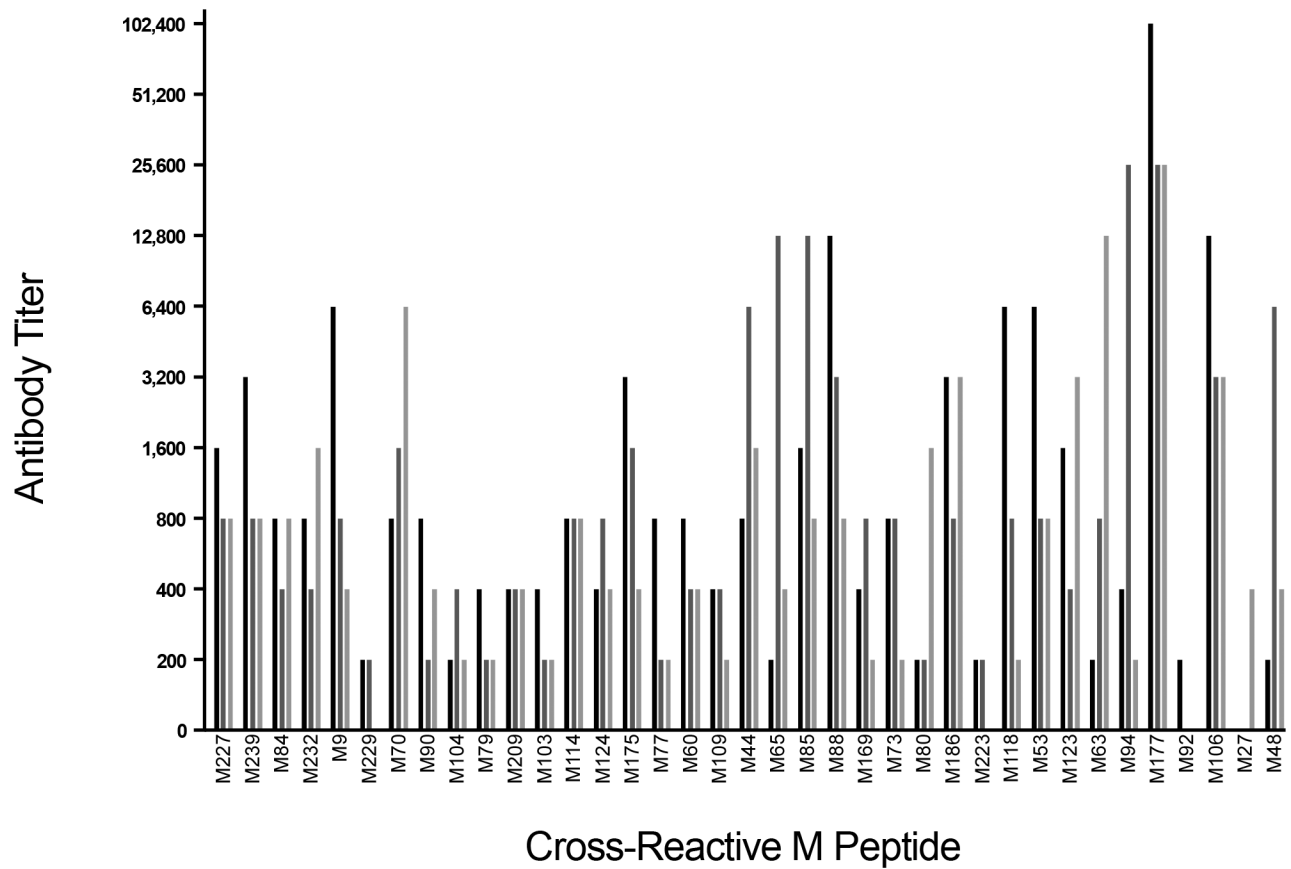


Fig. 2.

Serum antibodies elicited against M peptides contained in the 32-valent vaccine after immunization of three rabbits (each bar indicates the antibody titer for each individual rabbit). Pre-immune titers were ≤ 100 .

**Fig. 3.**

Serum antibodies elicited against M peptides predicted to cross-react with the 32-valent vaccine peptides after immunization of three rabbits (each bar indicates the antibody titer for each individual rabbit). Pre-immune titers were ≤ 100 .

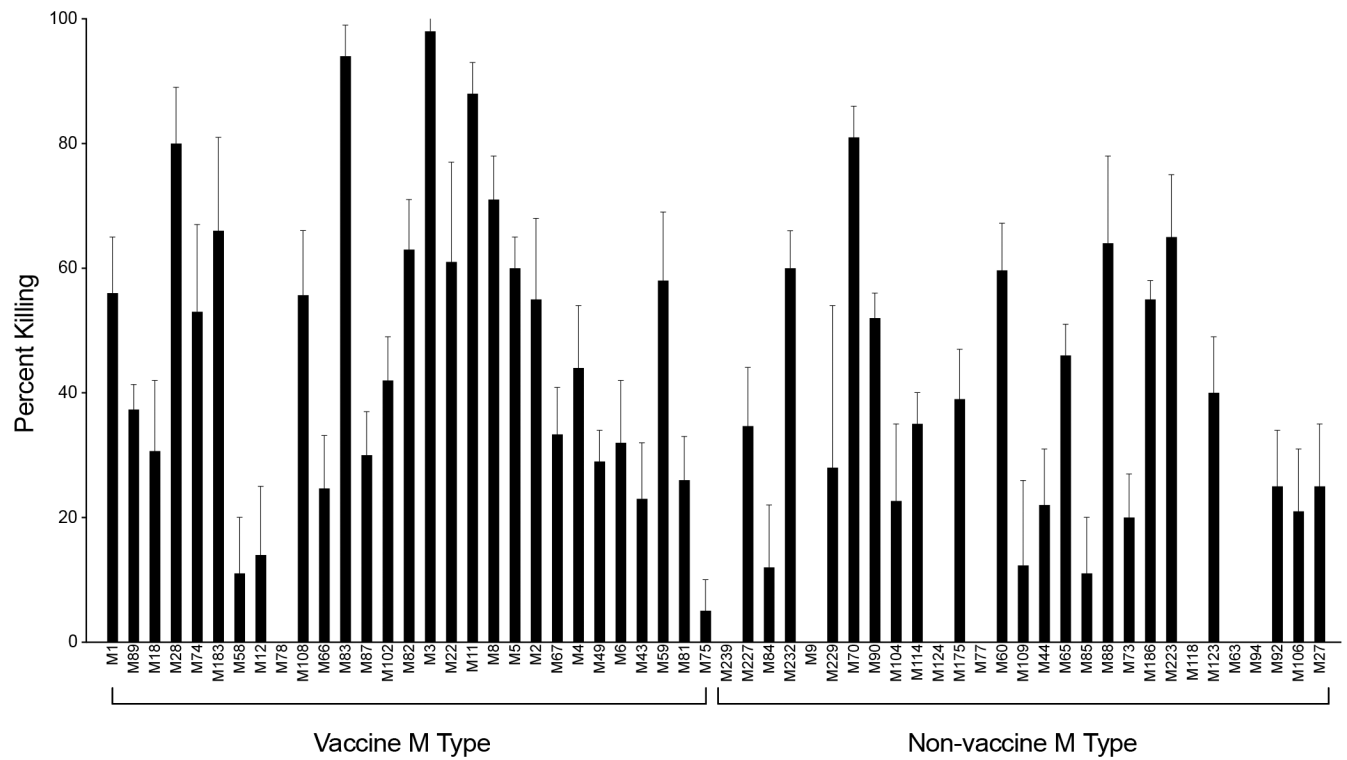


Fig. 4. Percent killing observed in HL-60 OPK assays of vaccine M types and non-vaccine M types of group A streptococci. Assays were performed in quadruplicate with immune serum from each rabbit and expressed as the mean $\pm$ S.D. The antiserum resulting in the highest mean killing against each M type is reported.

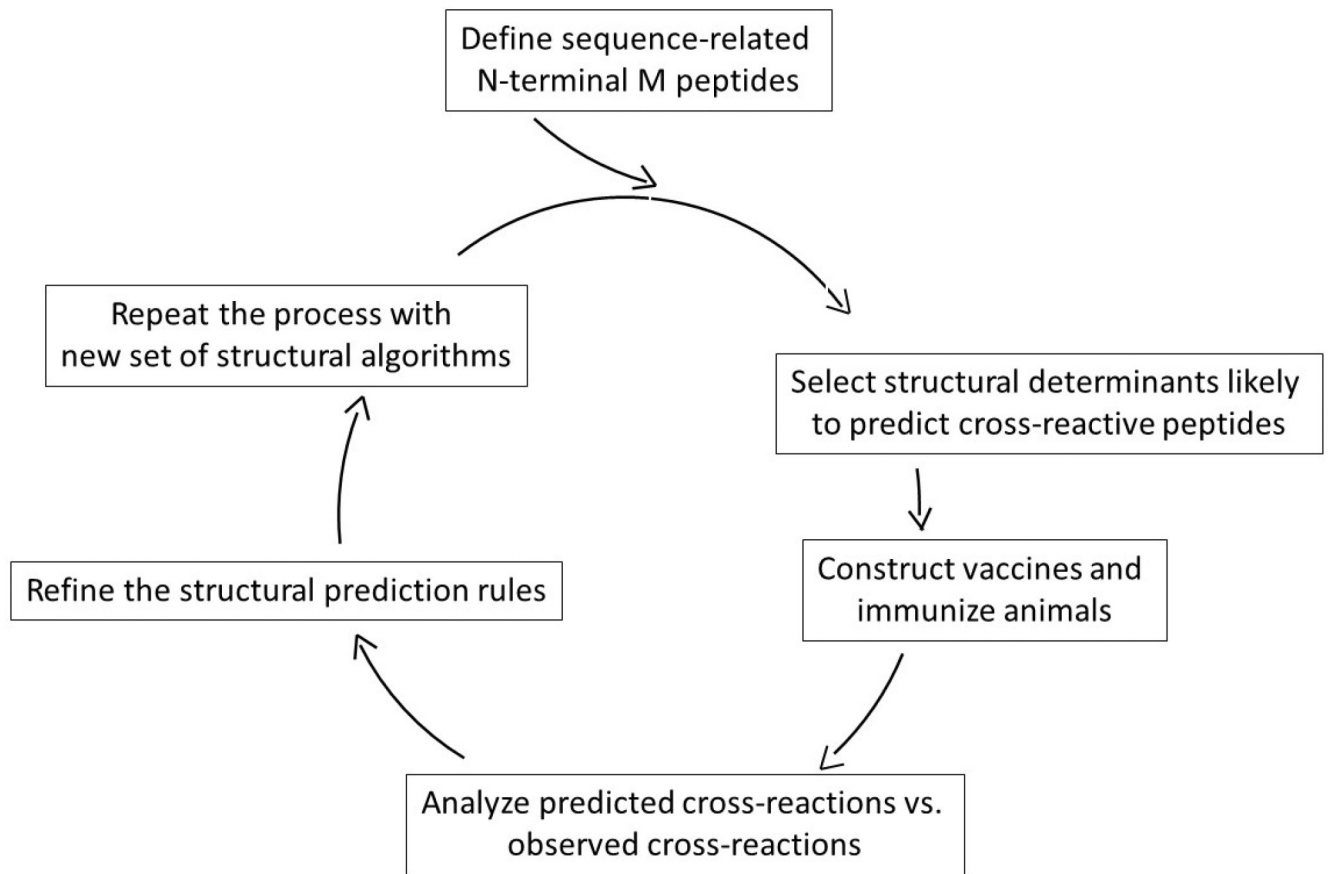


Fig. 5.
Iterative structure-guided design of multivalent M peptide vaccines.

Table 1.
Predicted and observed potential global and regional M type coverage¹ of 32-valent vaccine

	Africa	Arab States	Asia & Pacific	Europe	Oceania	North America	South America	World
Coverage based on predicted antibody responses and M type prevalence								
Vaccine M types	38.4	77.3	86.4	82.4	53.1	86.4	70.7	80.9
Cross-reactive M types	29.8	1.1	6.2	7.5	14.8	9.0	15.8	9.3
Total potential coverage	68.3	78.4	92.6	89.9	67.9	95.4	86.5	90.2
Coverage based on observed antibody responses and M type prevalence								
Vaccine M types	37.9	77.3	86.1	80.7	52.4	81.6	69.7	78.1
Cross-reactive M types	22.2	1.1	5.2	6.8	10.7	8.2	12.5	8.0
Total potential coverage	60.1	78.4	91.3	87.5	63.1	89.8	82.2	86.1

¹Coverage estimates are based on the M type prevalence of Strep A reported in [17].